

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

1. In hypothesis testing, the null hypothesis (H_0) typically represents:
 - a) The alternative explanation
 - b) The status quo or no effect
 - c) The hypothesis we aim to prove
 - d) The hypothesis with the most evidence
2. Which statistical test is appropriate for comparing the means of two independent groups?
 - a) Paired t-test
 - b) One-sample t-test
 - c) Independent samples t-test
 - d) Chi-square test
3. A confidence interval is a range of values that:
 - a) Gives the probability of observing a certain outcome
 - b) Contains the true population parameter with a certain level of confidence
 - c) Determines the significance of the result
 - d) Is used only in hypothesis testing
4. When the p-value is less than the significance level (α), what action should be taken regarding the null hypothesis?
 - a) Reject the null hypothesis
 - b) Fail to reject the null hypothesis
 - c) Accept the null hypothesis
 - d) None of the above
5. Which of the following factors affects the width of a confidence interval?
 - a) Sample size
 - b) Significance level
 - c) All of the above
6. The type II error in hypothesis testing is defined as:
 - a) Rejecting the null hypothesis when it is true
 - b) Failing to reject the null hypothesis when it is false

- c) Rejecting the alternative hypothesis when it is true
 - d) Accepting the null hypothesis when it is false
7. A 95% confidence interval for a population mean indicates:
- a) There is a 95% chance the population mean lies within the interval
 - b) There is a 5% chance the sample mean lies within the interval
 - c) There is a 95% chance the sample mean lies within the interval
 - d) There is a 5% chance the population mean lies within the interval
8. Which of the following is not a step in hypothesis testing?
- a) Formulate the null and alternative hypotheses
 - b) Collect and analyze data
 - c) Set the significance level after data analysis
 - d) Draw conclusions based on the data analysis
9. The margin of error in a confidence interval is affected by:
- a) Sample size and confidence level
 - b) Significance level and population size
 - c) Sample size and population size
 - d) Confidence level and population proportion
10. Statistical power in hypothesis testing is the probability of:
- a) Rejecting the null hypothesis when it is false
 - b) Failing to reject the null hypothesis when it is true
 - c) Rejecting the alternative hypothesis when it is true
 - d) Accepting the null hypothesis when it is false
11. Quality experts looked at a packaging process for ground beef over a series of days. An interesting question is whether the true mean amount delivered by this process changes from day to day. Ten packages were randomly selected. Ten more packages were separately randomly selected on another random day. The packages were all accurately weighed. The sample data is on Canvas labeled **Beef**.
- a. Is there evidence the average amount is changing between these two days? Test at level 0.05.
 - b. List the conditions you would need to check for this Case (thus, those not given in the problem), and check those that you can, otherwise say how you would check those assumptions.

12. It is suspected by researchers that there is less traffic on the streets on Friday the 13th than on other Friday's around the same time of year. To study this, they recorded the number of cars passing through two junctions in London on Friday the 6th and then the following Friday the 13th. They did this for all Friday the 13th's between July of 1990 and November of 1992. The sample data is on Canvas labeled **Friday13**.

- a. Test the appropriate hypothesis at level 0.10.
- b. List the conditions you would need to check for this Case (thus, those not given in the problem), and check those that you can, otherwise say how you would check those assumptions.

13. An engineer looked at nonconforming brick from a brick manufacturing process. Typically, 5% of the brick produced is not suitable for all purposes. Management monitors this process by periodically collecting random samples and classifying the bricks as conforming or nonconforming. A recent sample of bricks was taken, see data on Canvas **Bricks**.

- a. Is this evidence the percent of nonconforming bricks is less than the typical 5%?
- b. List the conditions you would need to check for this Case (thus, those not given in the problem), and check those that you can, otherwise say how you would check those assumptions.

14. Dole Pineapples, Inc. is concerned that the 16-ounce can of sliced pineapple is being overfilled. The quality-control department took a random sample of 50 cans. The data is on Canvas **Pineapples**. The population standard deviation is 0.1 ounces.

- a. Can you conclude that the mean weight is greater than 16 ounces? Test at level 0.05.
- b. List the conditions you would need to check for this Case (thus, those not given in the problem), and check those that you can, otherwise say how you would check those assumptions.

15. A sample of 15 college students is taken from XYZ University. Their GPAs were recorded. You can find the sample data on Canvas labeled **GPA**.

- a. Is there significant evidence to conclude the average GPA on this campus is greater than 3.0? Test at level 0.10.
- b. List the conditions you would need to check for this Case (thus, those not given in the problem), and check those that you can, otherwise say how you would check those assumptions.